ADDMC: Weighted Model Counting with Algebraic Decision Diagrams

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Problem: Boolean Model Counting

#SAT: computing number of satisfying assignments of Boolean formula

- Complexity: #P-complete [8]
- Example applications:
  - Medical diagnosis [7]
  - Power-transmission reliability analysis [3]

Weighted Model Counting

Boolean formula F and weight function W

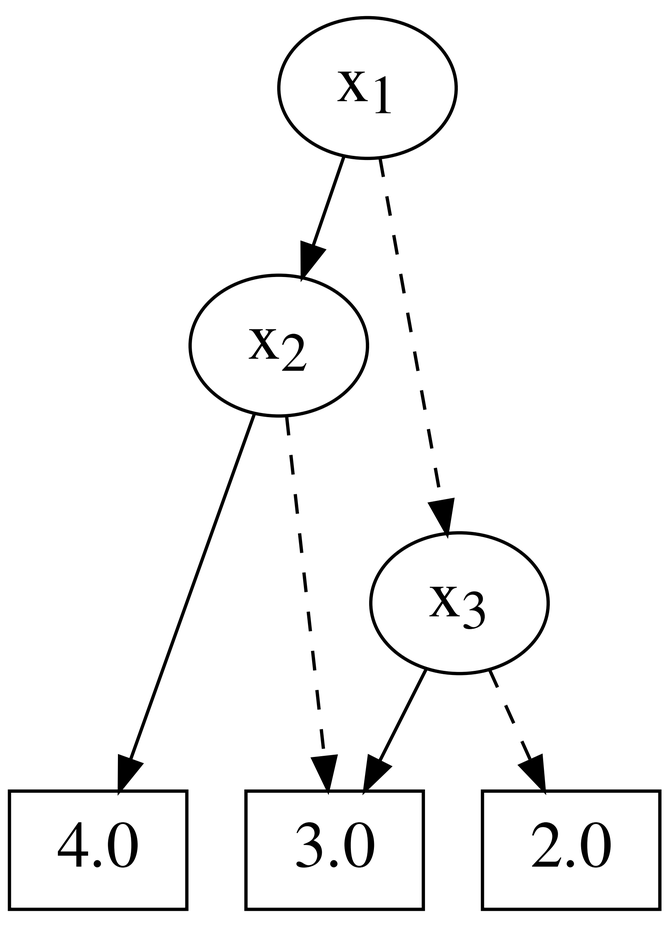
Assignment α			$F(\alpha)$	$W(\alpha)$	$(F \cdot W)(\alpha)$
x_1	x_2	x_3			
0	0	0	0	2.0	0.0
0	0	1	0	3.0	0.0
0	1	0	1	2.0	2.0
0	1	1	0	3.0	0.0
1	0	0	1	3.0	3.0
1	0	1	1	3.0	3.0
1	1	0	1	4.0	4.0
1	1	1	1	4.0	4.0

Weighted model count of F w.r.t. W

$$\#(F, W) = \sum_{\alpha \in 2^V} (F \cdot W)(\alpha) = 16.0$$

Algebraic Decision Diagram (ADD) [1]

Efficient data structure for weight function W[a]



[a] as well as for formula F and formula-weight product $F \cdot W$

Related Work on Model Counting

1. Search: exploring solution space in DPLL-like manner
   - Cachet [6]
2. Knowledge compilation: representing formula with efficient data structure *(exponential blowup in worst case)*
   - c2d [2]
   - miniC2D [5]
   - d4 [4]

Contributions to Model Counting

1. Algorithm for exact weighted model counting
   - Using efficient data structure: ADD
   - Exploiting factored representation of Boolean formula
   - Employing dynamic programming to combine ADDs of factors of formula *(mitigating exponential blowup)*
2. Implementation: ADD model counter (ADDMC)
   - Source code and experimental data publicly available[a]

[a] https://github.com/vardigroup/ADDMC

Dynamic Programming

Factored Representation

Conjunctive Normal Form (CNF) Boolean formula

$$\varphi = (x_1 \vee x_3) \wedge (\neg x_2 \vee x_3) \wedge (x_2 \vee \neg x_3) \wedge x_3$$

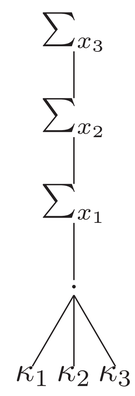
Clusters (partition of clauses)

$$\kappa_1 = \{x_1 \vee x_3\}$$
$$\kappa_2 = \{\neg x_2 \vee x_3, x_2 \vee \neg x_3\}$$
$$\kappa_3 = \{x_3\}$$

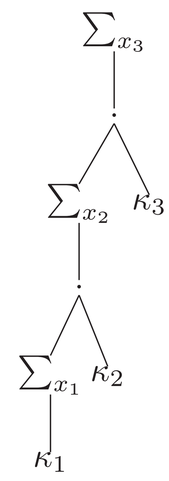
Early Projection

Reducing size of intermediate computation

Late projection



Early projection



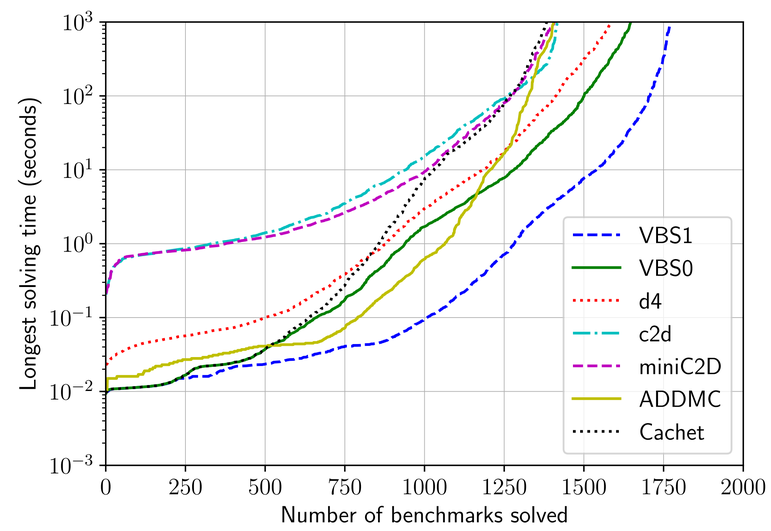
Experimental Benchmarks

1914 Boolean model counting benchmarks

- 1091 Bayesian networks[a]
- 823 instances from other domains[b]

[a] https://www.cs.rochester.edu/u/kautz/Cachet/
[b] http://www.cril.univ-artois.fr/KC/benchmarks.html

Experimental Results



Cactus plot of performance of two virtual best solvers (VBS1 with ADDMC; VBS0 without ADDMC) and five actual solvers

Acknowledgments

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References

[1] R Iris Bahar, Erica A Frohm, Charles M Gaona, Gary D Hachtel, Enrico Macii, Abelardo Pardo, and Fabio Somenzi. Algebraic decision diagrams and their applications. *Form Method Syst Des*, 10(2-3):171–206, 1997.

[2] Adnan Darwiche. New advances in compiling CNF into decomposable negation normal form. In *ECAI*, pages 328–332, 2004.

[3] Leonardo Duenas-Osorio, Kuldeep S Meel, Roger Paredes, and Moshe Y Vardi. Counting-based reliability estimation for power-transmission grids. In *AAAI*, 2017.

[4] Jean-Marie Lagniez and Pierre Marquis. An improved decision-DNNF compiler. In *IJCAI*, pages 667–673, 2017.

[5] Umut Oztok and Adnan Darwiche. A top-down compiler for sentential decision diagrams. In *IJCAI*, pages 3141–3148, 2015.

[6] Tian Sang, Fahiem Bacchus, Paul Beame, Henry A Kautz, and Toniann Pitassi. Combining component caching and clause learning for effective model counting. *SAT*, pages 20–28, 2004.

[7] Michael A Shwe, Blackford Middleton, David E Heckerman, Max Henrion, Eric J Horvitz, Harold P Lehmann, and Gregory F Cooper. Probabilistic diagnosis using a reformulation of the INTERNIST-1/QMR knowledge base. *Methods of Information in Medicine*, 1991.

[8] Leslie G Valiant. The complexity of enumeration and reliability problems. *SICOMP*, 8(3):410–421, 1979.